more herbivores and more predators. This is in contrast to top-down cascades, in which more predators lead to fewer herbivorous prey and a greater mix of plants.

“Just as a deer herd lives in mortal fear of its wolves, so does a mountain live in mortal fear of its deer.”

Aldo Leopold

Beetles, ants, and moths

Investigating trophic cascades in four-tier systems is more difficult, as predators at the top feeding level may eat predators at the level below and also herbivores below them, so the relationships become very complex. In 1999, researchers studying trophic cascades in tropical rain forest in Costa Rica got around this problem by studying a system of three trophic levels of invertebrates, in which the top predator—a clerid beetle—ate the predatory ants in the level below it, but not the herbivores in the level below that. When the number of predatory beetles in the study area was increased, the population of predatory ants fell dramatically. This reduced the pressure on dozens of species of herbivorous invertebrates, which therefore ate more vegetation. The leaf area of the plants in the study was consequently reduced by half.

Not all the “players” in trophic cascades are obvious or visible. Some are tiny and live underground. For example, yellow bush lupines—plants that live on the Californian coast—are consumed by the caterpillars of ghost moths, which eat the lupines’ roots. In turn, nematodes—wormlike invertebrates—parasitize the caterpillars. If these nematodes are in the soil, they will limit the population of caterpillars, and fewer of the lupines’ roots will be affected.